

Workshop on Essentials of Mortality Investigation of Asian Elephant

Detailed Workshop Notes

5th–6th June 2026

Dharamjaigarh & Raigarh Van Mandals
Bilaspur Forest Circle · Chhattisgarh

Organised by: **Chhattisgarh Forest Department**
Technical Support: **Wildlife Institute of India, Dehradun**
Laboratory Partner: **ICAR-IVRI, Izatnagar · NDVSU, Jabalpur**

1. Workshop Overview & Objectives

1.1 Context

This two-day residential workshop was convened by the Chhattisgarh Forest Department in partnership with the Wildlife Institute of India, Dehradun, to build technical capacity in elephant mortality investigation across the Bilaspur Forest Circle. The workshop brought together **84 participants** from the Forest Department and Veterinary Service Department of Chhattisgarh State.

1.2 Objectives

- **Biology, Behaviour & Disease:** Build a comprehensive understanding of Chhattisgarh elephant biology and behaviour, and the major infectious and non-infectious diseases causing mortality.
- **Necropsy, Biosafety & Sample Handling:** Develop hands-on competency in field necropsy under biosafety and biosecurity protocols.
- **Surveillance, Disposal & Remediation:** Establish operational frameworks for active and passive health surveillance, carcass disposal and site decontamination.

1.3 Expected Outcomes

- **Field Necropsy Skills:** Position and systematically eviscerate heavy elephant carcasses in remote field settings.
- **Sample & Lab Protocols:** Hands-on experience with tissue fixative formulation and secure packaging for histopathology, toxicology and microbiology.
- **Surveillance & Forensics:** Implementation of field-validated carcass disposal and site decontamination protocols, standardised forensic reporting, and proactive health monitoring frameworks.

2. Resource Persons

Dr. A. B. Shrivastav

Founder Director, School of Wildlife Forensic & Health; Professor & Veterinary Pathologist

Organisation	Nanjaji Deshmukh Veterinary Science University (NDVSVU), Jabalpur
Expertise	Wildlife Health Management · Wildlife Forensics · Veterinary Pathology

A renowned wildlife-health expert and founder Director of the School of Wildlife Forensic & Health at NDVSVU, Jabalpur (est. 2009). An architect of wildlife-health management in Central India since 1994, Prof. Shrivastav has advised national agencies including WII, NTCA, CZA, WWF and WTI, and supports forest departments across nine states. Trained at WII and the US National Wildlife Health Center, he has led 11 national projects — among them pioneering centres on wildlife forensics and the Indo-US Wildlife Health Co-operative Programme. Recipient of the Saheed Amrita Devi Vishnoi Award; 200 research papers, 60 Master's & PhD scholars.

Dr. Parag Nigam

Senior Scientist & Head, Wildlife Health Management

Organisation	Wildlife Institute of India, Dehradun
Expertise	Wildlife Health · Chemical Capture · Conservation Medicine

Veterinarian with nearly three decades of experience and former officer of the Indian Army's Remount & Veterinary Corps. Led India's first tiger reintroduction at Sariska and large-scale Indian Gaur reintroduction. Leading authority on chemical capture of wild animals; over 100 peer-reviewed publications spanning drug-protocol standardisation, disease ecology and One Health. Member, IUCN SSC Veterinary Specialist Group.

Dr. Karikalan Mathesh

Senior Scientist, Centre for Wildlife Conservation, Management & Disease Surveillance

Organisation	ICAR–Indian Veterinary Research Institute (IVRI), Bareilly
Expertise	Wildlife Pathology · Histopathology · Molecular Disease Diagnosis

Veterinary pathologist specialising in necropsy, histopathology, clinical pathology and molecular disease diagnosis in wild animals — particularly Asian elephants, wild carnivores and vultures. Led national research projects (ICAR, ICMR, DST-SERB, CZA, MoEF&CC). Over 150 peer-reviewed articles; Murray Fowler International Scholarship Award and Best Wildlife Pathologist Award (IAVP & AIZWV).

Dr. Tapendra Saini

Scientist C

Organisation	Wildlife Institute of India, Dehradun
Expertise	Animal Genetics & Breeding · Conservation Breeding · Wildlife Genetics

Veterinarian with BVSc & AH from SDAU, Gujarat and PG specialisation in Animal Genetics & Breeding from IVRI, Bareilly. Previously with Central Silk Board on tasar silkworm breeding in Saranda Forest, Jharkhand. Joined WII as Scientist C in April 2025, contributing to wildlife conservation-breeding programmes and disease-management initiatives.

Dr. Chandra Prakash Sharma

Principal Technical Officer & In-charge, Morphology Unit, WFCGC

Organisation	Wildlife Institute of India, Dehradun
Expertise	Wildlife Forensics · Morphological Identification · Crime-Scene Management

With WII since 1994; helped build the country's premier wildlife forensic facility through the WII–US Fish & Wildlife collaborative project (1995–2001). MSc Forensic Science (Punjabi University) and PhD Wildlife Science (Saurashtra University). Expert in morphological identification of illegal wildlife products, crime-scene management and evidence collection. Serves as expert scientific witness in courts nationwide; leads hands-on training and mock crime-scene exercises for law-enforcement officers across India.

3. Workshop Programme Schedule

The workshop spanned two days with 18 technical sessions covering anatomy, pathology, forensics, sampling and field practice.

Day 1 — 30 May 2026 (10:00–20:00+)

Time	Session	Faculty
10:00–11:00	Inauguration Official opening, workshop objectives, and introduction of experts & participants.	—
11:00–11:30	Elephant Status & Human-Elephant Interface in Chhattisgarh Population dynamics, corridor use, conflict and mortality trends.	PP, AM
11:30–12:15	Biological & Anatomical Aspects of Elephants Key anatomical peculiarities relevant to field investigation.	PN
12:15–13:00	Understanding Wildlife Mortalities: The Veterinary Perspective Carcass condition, natural vs unnatural death; ante/post-mortem changes & time of death.	PN
14:00–15:30	Infectious Causes of Mortality in Elephants Fatal infectious diseases: EEHV, EMC, FMD, Rabies, TB, Anthrax, Haemorrhagic Septicaemia.	KM, ABS
15:30–16:30	Non-Infectious & Disaster Mortalities in Elephants Electrocution, lightning, train hits, infighting, and other natural disaster deaths.	ABS, KM
16:30–17:30	Non-Infectious Diseases in Elephants Mycotoxicosis, kodo millet poisoning, and environmental bio-toxins.	KM, PN

Day 2 — 31 May 2026 (09:30–17:30)

Time	Session	Faculty
09:30–10:15	Biosafety & Biosecurity Measures in Elephant Mortality Disease threats from carcasses, essential PPE, safe disposal, site sanitation & disinfection.	KM, TS, PN
10:15–11:00	Equipment Familiarization & Site Preparation Heavy-duty necropsy equipment (chain saws, axes, winches) and establishing a secure perimeter.	ABS, KM, TS
11:30–12:30	Comprehensive Sampling Protocols Collection, packaging, cold chain, transport & documentation. Tissue selection for histopathology, toxicology & microbiology.	KM, ABS, PC, TS
12:30–13:00	Recording Information During Necropsy Report writing and photographic documentation standards.	KM, ABS
14:00–14:45	Vetro-Legal & Forensic Aspects of Mortality Investigation Investigation essentials for legal and forensic frameworks.	CP
14:45–16:00	Practical: Crime Scene Investigation Chain of custody, evidence collection, and legal procedures for ivory/tusk extraction.	CP, PN, AM, TS
16:15–16:45	Panel Discussion Critical aspects of managing mortalities in elephants.	All
16:45–17:30	Valedictory Workshop recap, feedback collection, and closing ceremony.	—

Faculty codes: ABS = Dr. A.B. Shrivastav · PN = Dr. Parag Nigam · KM = Dr. Karikalan Mathesh · TS = Dr. Tapendra Saini · CP = Dr. C.P. Sharma · PP = Paras Patel · AM = Asst. Manager

4. Elephant Status & Population — Chhattisgarh

4.1 Population Trend (2001–2026)

The elephant population in Chhattisgarh has grown dramatically over the past two decades, from a founding population of just 24 individuals in 2001 to an estimated **451 elephants** in 2026 — representing a compound annual growth rate (CAGR) of approximately 13%.

Year	Estimated Population	Notes
2001	24	Founding dispersal from Odisha/Jharkhand
2005	123	First major census count
2007	122	Stable
2015	247	Significant increase
2017	247	Stable — confirmed census
2021	279	Official Census
2026	~451 (est.)	Estimated; net +172 over 5-year study period

4.2 Key Population Facts

- **Bilaspur Circle** hosts the largest concentration of elephants in Chhattisgarh.
- **~10% CAGR** observed over the 2021–2026 study period (279 → 451).
- **2025-26** recorded the highest single-year human-elephant conflict events in the five-year study period.
- Population range spans **Dharamjaigarh** and **Raigarh Van Mandals**, with documented GPS tracking of herds and individual tuskers.

4.3 Live GPS Tracking

The presentation includes a live Leaflet.js-based map displaying real-time and recent GPS telemetry data:

- **17 GPS points** from herd elephants (shown in green).
- **29 GPS points** from individual tusker elephants (shown in amber).

The tracking system enables near-real-time monitoring of elephant movement corridors and early-warning alerts for human-elephant conflict zones.

5. Elephant Casualties — Five-Year Analysis (2021–2026)

5.1 Year-wise Deaths — Both Divisions

The five-year study period (2021-22 to 2025-26) with ongoing 2026-27 data recorded **49 confirmed elephant deaths** across Dharamjaigarh (DH) and Raigarh (RG) divisions.

Year	DH Division	RG Division	Total	Notes
2021-22	4	0	4	All DH
2022-23	6	6	12	RG emerges
2023-24	6	0	6	DH dominant
2024-25	6	4	10	Both divisions active
2025-26	1	11	12	RECORD year — RG peak
2026-27*	4	1	5	Ongoing — all calves, suspected drowning
TOTAL	27	22	49	5+ year study

■ **2025-26 is the record year with 12 deaths. The 2026-27 cohort (5 deaths in first 3 months — all calves, all suspected drowning) signals an acute emerging threat.**

5.2 Cause of Death Analysis

Analysis of all 49 confirmed deaths reveals electrocution as the dominant anthropogenic cause:

Cause of Death	Count	Percentage	Key Risk Factor
Electrocution	23	47%	Illegal / low-hung power lines near crop fields
Suspected Drowning	13	27%	Dams, ponds, seasonal nalas — primarily calves
Natural / Old Age	4	8%	Natural senescence
Fall / Trauma	4	8%	Hills, nalas, infighting
Other (incl. heat/sun)	5	10%	Heatstroke, weakness, undetrmined

5.3 Age Group Analysis

Age profiling reveals a disproportionate burden on the youngest individuals:

Age Class	Count	Percentage	Conservation Significance
Calf (0–2 yr)	23	50%	Highest single class — critical concern
Juvenile (2–15 yr)	5	11%	39% under 15 yr combined
Sub-adult (15–30 yr)	2	4%	Low
Adult (30–55 yr)	11	24%	Electrocution primary cause
Old (55+ yr)	5	11%	Female-skewed longevity

KEY CONCERN: 50% of all deaths are calves (0–2 yr). Combined with juveniles, 61% of deaths occur in individuals under 15 years — posing a direct long-term threat to population viability.

5.4 Sex Profiling (45 confirmed records)

Sex	Count	Distribution	Leading Causes
Male	23	52%	Electrocution (11), Suspected Drowning (6), Falls (3)
Female	22	49%	Electrocution (7), Suspected Drowning (5), Natural/Old age (3)

Critical Sex-Specific Patterns:

- **Calf Vulnerability (12F vs 10M):** 55% of calf deaths are female. Raigarh: 8 of 11 drowning deaths are young females (6 months–2 years).
- **Electrocution (11M, 7F):** Males at higher risk — 48% of male deaths from electrocution vs 33% of females.
- **Suspected Drowning (6M, 5F):** Female-skewed among calves — 5 of 6 drowning females are calves.
- **Old Age Deaths (1M, 4F):** Female longevity evident — more females survive to 55+ years.

5.5 Seasonal Variation

Monthly analysis reveals two distinct mortality peaks:

Month	Total Deaths	Drowning	Electrocution
January	6	3	2
February	4	0	0
March	3	1	2
April	2	0	1
May	5	4	0
June	5	1	1
July	1	0	1
August	1	0	1
September	3	0	2
October	8	1	7
November	6	2	1
December	5	2	2

Key Seasonal Patterns:

- **October peak (8 deaths):** Post-monsoon movement & harvest season overlap. Oct–Dec = 42% of annual deaths.
- **Electrocution — Oct–Dec dominates (10 deaths, 50%):** Electric fencing around standing crops is most active during this post-monsoon harvest window.
- **Drowning — Jan–May dominates (8 deaths, 62%):** Dry season — pond banks are steep and water levels low, trapping young calves that fall in while drinking.

5.6 Range-wise Distribution

Casualty hotspots by forest range/beat:

Range / Beat	Division	Deaths	Risk Level	Elec	Drown	Other
Ghargoda	Raigarh	15	CRITICAL	7	5	3
Chhal	Dharamjaigarh	14	CRITICAL	5	6	3

Dharamjaigarh HQ	Dharamjaigarh	7	HIGH	5	0	2
Tamanar	Raigarh	5	MODERATE	4	1	0
Borojh	Dharamjaigarh	4	HIGH	1	0	3
Lailungaan	Dharamjaigarh	2	HIGH	0	0	2
Baakaaruma	Dharamjaigarh	1	MODERATE	1	0	0
Raigarh HQ	Raigarh	1	LOW	0	1	0
Kharsiya	Raigarh	1	LOW	0	1	0

IMMEDIATE ACTION REQUIRED: Chhal (DH) & Ghargoda (RG) together account for 29 deaths — 59% of all casualties. These two ranges require priority electrocution and drowning mitigation interventions.

6. Suspected Drowning Cases — Detailed Analysis

6.1 Overview

Suspected drowning is the second-leading cause of elephant mortality in the study area, accounting for **13 deaths (27%)**. Notably, **92% of drowning victims are calves or young individuals**, making this pattern the most acute threat to population recruitment and long-term viability.

6.2 Gross Post-mortem Findings — All 13 Cases

The following PM findings were consistent across all or most suspected drowning cases:

PM Finding	Frequency
Frothy nasal/oral discharge — emphysema aquosum	ALL CASES
Lungs — waterlogged, heavy, congested, oedematous	ALL CASES
Water in trachea & bronchi	ALL CASES
Gastric content — large volume of water & plant debris	ALL CASES
Skin maceration / hide wrinkling	CONSISTENT
Petechial haemorrhages — conjunctiva / sub-pleural	VARIABLE

6.3 Histopathology — Microscopic Confirmation

- Pulmonary oedema — alveolar flooding with eosinophilic fluid.
- Pulmonary haemorrhage — interstitial and alveolar spaces.
- Hepatocyte degeneration — confirmed in 3 cases (Tamnar range, December 2025).
- **Diatom test positive** — confirmatory for ante-mortem drowning in all tested cases.
- Renal congestion — passive vascular engorgement.

6.4 ARSENIC DETECTION — Critical Finding

ARSENIC DETECTED
ICAR-IVRI Lab Report · Tamnar Range, December 2025

Arsenic detected at elevated levels in visceral samples from **2 specimens in Raigarh Division** (Tamnar Range, December 2025). Histopathology confirmed hepatocyte degeneration. Both cases classified as suspected drowning — arsenic likely a contributing metabolic stress factor from contaminated dam water near industrial corridor. **Environmental sampling of Rabo / Panikshet dam catchments recommended immediately.**

6.5 Toxicology — Negative Findings

All tested samples (5 batches) returned **NEGATIVE** for the following:

- EEHV (Elephant Endotheliotropic Herpesvirus) — Negative
- Nitrate-Nitrite compounds — Negative
- Organophosphate insecticides — Negative
- Heavy metals (Pb, Hg, Cd) — Negative
- Hydrogen Cyanide (HCN) — Negative

This clears the most common pesticide and industrial toxin hypotheses, narrowing the causal mechanism to drowning with possible arsenic co-exposure.

7. Post-mortem Report — Summary Analysis

7.1 Lead Cause — PM Confirmed

Electrocution leading to Cardio-Respiratory Failure is confirmed as the cause of death in **20 records (49%)** of PM-confirmed cases. Multiple simultaneous electrocution events were noted in 3 cases. Suspected Drowning is the second-leading confirmed cause (9 records, 22%).

7.2 PM Cause Distribution

Cause	PM-Confirmed Count	Percentage
Electrocution	~20	49%
Suspected Drowning	~9	22%
Natural / Old Age	~5	12%
Trauma / Fall	~5	12%
Other	~2	5%

7.3 Age Concern — Neonates & Calves

Calves (0–2 yr) account for **49% of all age-confirmed records** (22 of 45). This level of early juvenile mortality poses a **direct threat to long-term population viability**. The combination of drowning risk in young calves during the dry season and arsenic contamination in water bodies near industrial corridors requires immediate field intervention.

8. Laboratory Report Analysis

8.1 ICAR-IVRI & NDVSU Reports

Laboratory analyses were conducted by **ICAR-IVRI, Izatnagar** (File F.2-19/DI/NRC/2025-26/CWL) and the **School of Wildlife Forensic & Health, NDVSU, Jabalpur** across five tested batches.

8.2 Arsenic Detection

Two specimens from Tamnar Range, Raigarh Division (December 2025) tested positive for **arsenic at elevated levels** in visceral samples. Histopathology confirmed hepatocyte degeneration — suggesting chronic or sub-acute arsenic poisoning likely from contaminated dam water near the industrial corridor. Environmental sampling of the Rabo and Panikshet dam catchments is recommended immediately.

8.3 EEHV — Negative

EEHV-I (Elephant Endotheliotropic Herpesvirus) PCR testing returned **NEGATIVE** in all tested samples, confirmed independently by NDVSU Jabalpur for Dharamjaigarh Division specimens. No viral haemorrhagic disease involvement — viral aetiology is ruled out. This is particularly significant given the high neonate mortality observed.

8.4 Broad Toxicological Screen — Negative

Five tested batches returned **NEGATIVE** results for: Nitrate-Nitrite compounds, Heavy metals (Pb, Hg, Cd), Organochlorine & organophosphate insecticides, Hydrogen Cyanide (HCN). The most common pesticide and industrial toxin hypotheses are cleared.

9. Case Study — Gurda Elephant Casualty (01 June 2026)

9.1 Incident Details

Date	01 June 2026
Location	Gurda, Kharsia Range
Division	Raigarh Van Mandal
GPS Coordinates	22.0439°N, 83.1313°E (22°3′16″N 83°8′11″E)
Elevation	256.04 ± 9.59 m
Water Body	Gurda River / Sandbank

9.2 PM Report Summary

Species	Asian Elephant (<i>Elephas maximus</i>)
Age Class	Calf (0–2 yr)
Cause of Death	Suspected Drowning
Body Condition	Found on exposed riverbank sandbank
Pattern	Consistent with dry-season drowning — steep bank, low water, young calf

9.3 Field Notes

The calf was found on an exposed sandbank adjacent to the Gurda River water body in Kharsia Range, Raigarh Van Mandal. The incident is consistent with the established dry-season drowning pattern — when water levels are low and pond/river banks are steep, young calves that approach the water edge to drink can fall in and are unable to climb out. Field photographic documentation was secured at 10:56 and 11:03 hrs on 06-01-2026 from aerial and ground perspectives respectively.

Field note (original): ■■■■ ■■■■■■■■ ■■■■ ■■■■■■ — Gurda ■■■■■■■■■■, ■■■■■■ ■■■■■■■■ ■■■■■■■■

This case represents the 2026-27 cohort pattern — all 5 deaths in the current year are calves, all suspected drowning, all in the first 3 months of the year.

10. Key Findings, Recommendations & Action Points

10.1 Priority Intervention Zones

- **CRITICAL:** Chhal range (DH) and Ghargoda range (RG) — combined 29 deaths (59%). Immediate multi-agency intervention required.
- **HIGH:** Dharamjaigarh HQ and Borojh (DH) — electrocution dominant. Line patrolling and insulator replacement urgently needed.
- **MODERATE:** Tamarar (RG) — mixed electrocution and drowning; seasonal watch required Oct–Dec.

10.2 Electrocution Mitigation

- Survey and replacement of **low-hung power lines** in all elephant corridors, particularly within 1 km of identified hotspot ranges.
- Seasonal enforcement campaign targeting **Oct–Dec** (crop harvest period) when illegal electric fencing is most active.
- **Multi-simultaneous electrocution** events noted in 3 cases — indicates persistent illegal installations; criminal liability must be enforced.

10.3 Drowning Prevention

- Installation of **earthen ramps / exit points** at high-risk dam edges and ponds within confirmed elephant drinking zones.
- **Fencing or barriers** at steep-banked water bodies identified as drowning sites (particularly Panikshet and Rabo dams, Raigarh).
- Priority focus on Jan–May period — 62% of drownings occur in this window.
- Real-time calf monitoring in herds using GPS data during dry season.

10.4 Environmental Sampling

IMMEDIATE: Environmental sampling of Rabo & Panikshet dam catchments, Raigarh Division.

Arsenic detected in 2 December 2025 specimens near industrial corridor. Water and sediment sampling required; results to be shared with Chhattisgarh Pollution Control Board for enforcement action.

10.5 Surveillance & Reporting Enhancements

- Standardised **Proforma-1 documentation** to be used consistently across all ranges — including GPS coordinates, age estimate, cause, and photo log.
- All mortality events to trigger **mandatory PM within 24 hours** and sample dispatch to ICAR-IVRI within 48 hours with cold chain documentation.
- Quarterly mortality review meetings at division level with range officers.
- EEHV surveillance to continue annually given high neonate mortality — negative results must be systematically documented.

10.6 Summary Statistics

Metric	Value
Study Period	2021-22 to 2026-27 (ongoing)

Total Elephant Deaths	49 confirmed
Divisions Covered	Dharamjaigarh (27) + Raigarh (22)
Leading Cause	Electrocution — 23 deaths (47%)
Second Cause	Suspected Drowning — 13 deaths (27%)
Most Vulnerable Age	Calves 0–2 yr — 23 deaths (50%)
Deadliest Year	2025-26 — 12 deaths (record)
2026-27 Trend	5 deaths in first 3 months — all calves
Critical Hotspot Ranges	Chhal (14) + Ghargoda (15) = 59% of all deaths
Drowning Seasonality	Jan–May = 62% of drowning deaths
Electrocution Seasonality	Oct–Dec = 50% of electrocution deaths
Arsenic Detected	2 specimens — Tamnar Range, Dec 2025
EEHV Status	Negative (all batches)
Toxicology	Negative for all major toxins screened
Workshop Participants	84 (Forest Dept + Veterinary Service Dept, CG)

Data Sources: Official Proforma-1 Records, Dharamjaigarh & Raigarh Divisions · PM Reports: ICAR-IVRI Izatnagar (File F.2-19/DI/NRC/2025-26/CWL) · GIS Data: Bilaspur Forest Division GIS Cell · Generated: 30 May 2026

Organised by: Chhattisgarh Forest Department · Technical Support: Wildlife Institute of India, Dehradun · Laboratory: ICAR-IVRI, Izatnagar & NDVSU, Jabalpur